

Notes: Edmans, Goldstein, and Jiang (JF, 2012)
The Real Effects of Financial Markets: The Impact of Prices on Takeovers

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1 Big picture

This paper asks whether financial market prices have **real effects** on corporate control. The central question is simple: does a low stock-market valuation make a firm more likely to receive a takeover bid?

The paper argues that the observed relation between valuation and takeover likelihood is hard to interpret because of two opposite forces:

- **Trigger effect:** a lower valuation can create a profit opportunity for acquirers and therefore increase takeover probability.
- **Anticipation effect:** if the market expects a future takeover, current prices rise today, which reduces the measured valuation discount.

The empirical challenge is that these two forces operate simultaneously. A naive regression of takeover probability on valuation may find a weak relation even if prices strongly affect takeovers, because prices themselves already incorporate expected takeovers.

Main result. After instrumenting valuation shocks with mutual-fund-flow-induced price pressure, an interquartile increase in the valuation discount raises takeover probability by about **5.7–7.6 percentage points**, compared with an unconditional takeover probability of about **6.2%**. This is much larger than the non-instrumented relation.

Interpretation. The market is not just a mirror of future fundamentals. Prices can change real corporate outcomes. In this paper, temporary nonfundamental price declines can make firms more attractive takeover targets.

2 Incremental contribution of the paper

The paper contributes in five ways.

1. **Direct causal evidence on real effects of financial markets.** Earlier work, such as studies of investment-price sensitivity, often shows that real decisions are more sensitive to prices when prices are more informative. This paper uses an instrument for price changes and directly estimates the causal effect of market prices on a major real corporate event: takeovers.
2. **Separates the trigger effect from the anticipation effect.** Prior takeover studies often find only weak links between valuation and takeover probability. This paper shows that the weak relation can arise because expected takeovers raise prices before the bid occurs.
3. **Introduces a valuation discount measure rather than raw valuation.** The key variable is not simply Tobin's Q or $EV/Ebitda$. It is the gap between actual valuation and an estimated potential or frontier valuation.
4. **Uses mutual fund redemptions as a source of nonfundamental price pressure.** Mutual fund outflows mechanically force funds to sell stocks they already hold. This creates temporary price pressure that is plausibly unrelated to the target-specific likelihood of takeover.
5. **Shows a feedback loop between markets and corporate events.** Prices affect takeover likelihood, and takeover likelihood affects prices. This two-way relation is central to understanding why simple correlations are misleading.

Incremental message. The paper moves the literature from “prices predict takeovers” to “prices can cause takeovers.” It also shows that forward-looking prices can be a double-edged sword: they discipline managers by triggering takeovers, but anticipation of takeovers can raise prices and partially deter the very takeover threat that the market expects.

3 Core economic mechanism

3.1 Why low valuation can trigger takeover bids

A firm is attractive to an acquirer when its market price is low relative to the value that the acquirer believes can be created under new ownership or management. The acquirer may profit if the purchase price is tied to the current market price but the acquirer can improve the firm after acquisition.

Let V_{it} be the observed valuation of firm i and V_{it}^* be its potential or frontier valuation. The paper defines a discount as

$$\text{Discount}_{it} = \frac{V_{it}^* - V_{it}}{V_{it}^*}.$$

A high discount means that the firm trades far below estimated potential value. The trigger hypothesis predicts that higher discount increases takeover probability.

3.2 Why the same market price also reflects takeover anticipation

If investors expect a takeover, they anticipate a takeover premium. This expectation pushes the current stock price upward. Therefore, takeover probability affects the measured discount even before any bid is announced.

The paper models this feedback using a simultaneous system:

$$\text{Discount}_{it} = \gamma_0 X_{it} + \gamma_1 Z_{1,it} + \gamma_2 Z_{2,it} + \delta \xi_{it} + \eta'_{it},$$

$$\text{Takeover}_{it}^* = \mu_1 \text{Discount}_{it} + \mu_2 X_{it} + \mu_3 Z_{1,it} + \xi_{it}.$$

Here μ_1 is the **trigger effect**: how much the valuation discount raises takeover probability. The term $\delta \xi_{it}$ captures the feedback from takeover likelihood to the discount. A negative δ means that higher takeover likelihood lowers the discount because prices rise in anticipation.

Interpretation. Without an instrument, estimates of μ_1 are biased downward because the anticipation effect makes high-takeover-likelihood firms look less discounted.

4 Data, measurement, and variables

4.1 Takeover data

The paper uses Securities Data Company (SDC) takeover data from 1980 to 2007. It includes bids, not only completed acquisitions, because target valuation is expected to matter most for the acquirer’s decision to make a bid. The sample excludes financial firms and utilities because takeovers are highly regulated in these sectors.

The dependent variable is

$$\text{Takeover}_{it} = 1$$

if firm i receives a takeover bid in a calendar year, and zero otherwise. The full sample contains **6,555 deals**. The universe of potential targets is nonfinancial, nonutility firms with necessary CRSP and Compustat data.

4.2 Valuation discount

The paper constructs the discount in two ways.

Industry frontier. The potential value is the 80th percentile valuation among firms in the same three-digit SIC industry. This approach is simple and close to practitioners' comparable-company analysis.

Firm-characteristic frontier. The potential value is estimated using firm characteristics that acquirers are unlikely to change immediately, including age, size rank, asset turnover, R&D intensity, market share, growth, and asset beta.

Two valuation metrics are used:

- Q , the ratio of enterprise value to book value;
- EV/Ebitda, the ratio of enterprise value to cash-flow-like operating earnings.

This produces four discount measures: industry- Q , firm- Q , industry-EV/Ebitda, and firm-EV/Ebitda.

4.3 Instrument: mutual fund price pressure

The key instrument is **MFFlow**, mutual-fund-flow-induced price pressure. The logic is as follows:

- Mutual fund investors redeem shares for reasons unlikely to be driven by takeover prospects of individual firms held by the fund.
- Funds experiencing large outflows must sell existing holdings.
- This selling pressure temporarily depresses prices.
- The price pressure affects takeover likelihood only through its effect on market valuation.

The paper excludes sector funds and controls for fundamentals and year effects to reduce the concern that fund flows directly reflect industry takeover waves.

Interpretation. MFFlow is powerful because it isolates nonfundamental price movements. This allows the authors to identify whether prices themselves affect takeover activity.

5 Identification and empirical design

The identification strategy has two parts.

First, estimate valuation discounts. The authors estimate potential value using stochastic frontier logic, implemented with quantile regression. The discount is the gap between actual and potential value.

Second, instrument the discount. The observed discount is endogenous because of anticipation, omitted variables, and measurement error. Mutual fund outflow-induced price pressure provides an excluded source of variation in market prices.

The main empirical comparison is:

- **Without feedback/instrumentation:** how discount and takeover probability are correlated in equilibrium.
- **With feedback/instrumentation:** how an exogenous discount shock affects takeover probability.

Interpretation. The paper’s empirical design is useful because it shows why the equilibrium relation between valuation and takeover probability understates the causal effect of prices.

6 Main findings

- **Discount predicts takeovers even without feedback, but the magnitude is small.** An interquartile change in discount raises takeover probability by roughly 0.4–1.6 percentage points depending on the discount measure.
- **After correcting for feedback, the trigger effect is much larger.** An interquartile increase in discount raises takeover probability by roughly 5.7–7.6 percentage points.
- **Takeover probability feeds back into prices.** A one percentage point increase in takeover probability reduces the measured discount by roughly 1.2 percentage points for the average firm.
- **MFFlow creates temporary price pressure.** Stocks subject to large mutual fund outflows fall sharply and recover over roughly two years, supporting the instrument’s interpretation as temporary, nonfundamental price pressure.
- **The results are robust for financially driven takeovers.** The trigger effect remains strong for leveraged buyouts and acquisitions by financial sponsors.

7 Extracted figures and tables from the paper

7.1 Variable construction and sample description

Table I Summary of Variables									
This table summarizes the main variables used (Panel A) and presents summary statistics (Panel B). All data are obtained from Compustat unless otherwise stated; "data" numbers refer to the line items from Compustat.									
Panel A: Data Definitions									
Discount Variables (<i>Discount</i>)									
Discount (Industry: EV/Ebitda)	Value discount relative to the industry frontier, using EV/Ebitda as the valuation metric.								
Discount (Industry: Q)	Value discount relative to the industry frontier, using Q as the valuation metric.								
Discount (Firm: EV/Ebitda)	Value discount relative to the firm-specific frontier, using EV/Ebitda as the valuation metric.								
Discount (Firm: Q)	Value discount relative to the firm-specific frontier, using Q as the valuation metric.								
Fundamental Variables (<i>X</i>)									
Age	Firm age, calculated as years from first appearance in CRSP.								
ATO	Asset turnover. Sales (data12)/Assets (data6).								
BetaAsset	Beta on the market factor in a Fama-French three-factor model using daily data from CRSP, and then unlevered.								
Growth	Average sales growth during past (up to) 3 years.								
MktShr	Sales / Total sales in three-digit SIC industry.								
R&D	R&D expense (data46)/Sales (data12); zero if missing.								
SalesRank	Rank of sales (data12) among all Compustat firms in a given year, ranging from zero to one.								
Variables Affecting Discount and Takeover Probability (<i>Z₁</i>)									
Amihud	Illiquidity measure per Amihud (2002); yearly average of the square root of (Price × Vol)/Return.								
HHIFirm	Herfindahl index of a firm's sales in different business segments.								
HHISIC3	Herfindahl index of sales by all firms in three-digit SIC industry.								
Inst	% of shares outstanding held by institutions, from Thomson Financial.								
Leverage	(Debt (data9 + data34) – Cash (data1))/Assets (data 6).								
Payout	(Dividends (data21) + Repurchases (data115))/Net Income (data18); zero if numerator is zero or missing, and one if numerator > 0 and denominator = 0.								
Variables Affecting Discount (<i>Z₂</i>)									
MFFlow	Mutual fund price pressure, from Thomson Reuters mutual fund holdings database. See the Appendix for further details.								
Panel B: Summary Statistics									
Name	# Obs	Mean	Std. Dev.	Percentiles					
				5 th	25 th	50 th	75 th	95 th	
Age	118,942	11.48	13.03	1	3	7	15	37	
ATO	118,942	1.21	0.82	0.17	0.63	1.08	1.59	2.79	
Amihud	101,026	0.77	1.11	0.02	0.11	0.35	0.93	3.05	

(continued)

Table I—Continued									
Panel B: Summary Statistics									
Name	# Obs	Mean	Std. Dev.	Percentiles					
				5 th	25 th	50 th	75 th	95 th	
BetaAssets	117,211	0.69	0.41	0.09	0.38	0.65	0.95	1.45	
Discount (Industry: EV/Ebitda)	92,116	0.18	0.48	−1.05	0.10	0.38	0.57	0.76	
Discount (Industry: Q)	116,543	0.24	0.47	−0.90	0.09	0.37	0.57	0.77	
Discount (Firm: EV/Ebitda)	92,141	0.27	0.48	−1.03	0.11	0.41	0.61	0.79	
Discount (Firm: Q)	116,567	0.28	0.46	−0.92	0.11	0.41	0.60	0.77	
EV/Ebitda	92,141	15.95	28.05	3.76	6.12	8.70	13.77	47.05	
Growth (%)	118,942	30.4%	80.0%	−17.8%	1.3%	11.4%	28.3%	127.5%	
HHIFirm	118,942	0.85	0.24	0.35	0.66	1.00	1.00	1.00	
HHISIC3	118,942	0.19	0.16	0.06	0.09	0.14	0.25	0.50	
Inst (%)	118,942	27.9%	26.7%	0.0%	4.1%	19.8%	46.8%	80.4%	
Leverage (%)	118,942	8.8%	34.6%	−56.5%	−11.7%	12.5%	31.8%	60.5%	
MFFlow	118,942	−0.30	0.92	−1.45	−0.20	0.00	0.00	0.00	
MktShr (%)	118,942	5.1%	12.8%	0.0%	0.1%	0.5%	3.3%	27.4%	
Payout (%)	118,942	38.1%	77.4%	0.0%	0.0%	0.0%	50.3%	137.0%	
Q	116,567	2.33	2.55	0.67	1.04	1.51	2.51	6.75	
R&D (%)	118,942	19.0%	114.4%	0.0%	0.0%	0.0%	4.7%	38.2%	

Table I, continued: summary statistics.

Table I, first part: variable definitions and beginning of summary statistics.

Read:

- The paper's key variable is *Discount*, measured relative to either an industry frontier or a firm-specific frontier.
- The mean discount ranges from about 18% to 28%, depending on the valuation metric and frontier definition.
- MFFlow is the mutual-fund-price-pressure instrument, constructed from fund outflows and existing fund holdings.
- The sample contains more than 100,000 firm-year observations for the main *Q*-based measures and about 79,000 observations for EV/Ebitda-based specifications.

Economic message: The paper does not rely on raw valuation. It builds a discount-to-potential-value measure that is closer to the takeover logic: acquirers should care about whether a firm is cheap relative to what it could be worth, not simply whether its raw *Q* is low.

7.2 Aggregate patterns and the mutual fund price-pressure event study

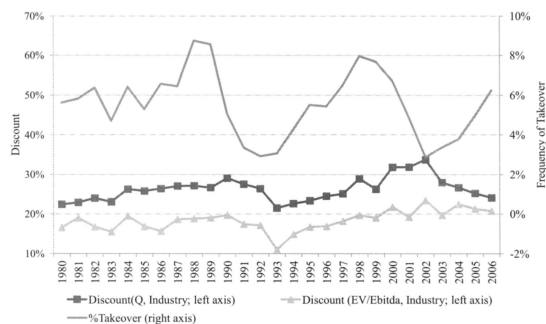


Figure 1. Time series of aggregate discounts and takeover activities. This figure plots the time series (1980 to 2006) of the aggregate *Discount* (as defined in Table 1, Panel A, the left axis)

Figure 1: aggregate discounts and takeover activity.

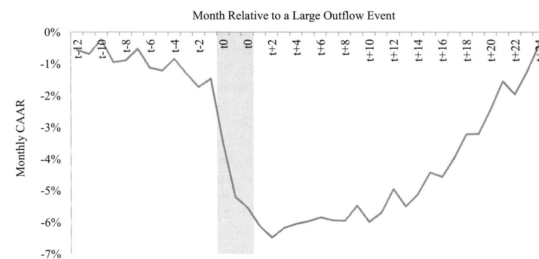


Figure 2. Effect of mutual fund outflows on stock returns. This figure plots the monthly cumulative average abnormal returns (CAAR) of stocks around the event months, where an event is defined as a firm-month observation in which *MFFlow* falls below the 10th percentile value of the full sample. CAAR is computed over the benchmark of the CRSP equal-weighted index from 12 months before the event to 24 months after. (There are three event months because holdings are only recorded at the quarterly level, while returns are recorded at a monthly frequency.)

Figure 2: stock returns around large mutual fund outflow events.

Read:

- Figure 1 plots aggregate discount measures and the frequency of takeovers over 1980–2006.
- Discount and takeover activity tend to move together, but this aggregate relation is not enough to identify causality.
- Figure 2 shows that stocks hit by large mutual fund outflow pressure fall substantially around the event month.
- The price decline is persistent but eventually reverses, with recovery by about 24 months after the event.

Economic message: Figure 2 is central for the instrument. It shows that MFFlow produces a temporary price shock rather than a permanent deterioration in fundamentals. This is why it can be used to isolate the real effect of price changes on takeover bids.

7.3 Baseline estimates without modeling feedback

Table II
Determinants of Discount and Takeover without Feedback

This table reports results from estimating equations (4) and (5) separately. All variables are defined in Table I. The dependent variable in Panel A is *Takeover*, and that in Panel B is *Discount*. The firm-specific frontier is formed by a quantile (at the 80th percentile) regression of valuation measures on *SalesRank*, *R&D*, *DTT*, *MarketShare*, *Growth*, *ReturnAssets* (all expressed in terms ranks), *Age*, and *Age²*. Year fixed effects are used in all specifications but are unreported. All standard errors are adjusted for heteroskedasticity and within-cluster correlation. In Panel A, they are clustered at the firm level; in Panel B, they are double-clustered at the year and firm levels, and the corresponding *t*-statistics are reported in the brackets below the coefficients. The column *dP/dX* gives the marginal effect on takeover probability of a one unit (or 100 percentage point) change in each regressor. The bottom row of the table reports the number of observations, the pseudo R^2 , and the all-sample frequency of the dependent variable being one. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels.

Panel A: Determinants of Takeover

	Dependent Variable = Takeover									
	Discount = Discount(Q)				Discount = Discount(EV/Ebids)					
	Industry-Specific Frontier		Firm-Specific Frontier		Industry-Specific Frontier		Firm-Specific Frontier			
	Coef	<i>t</i> -Stat	dP/dX	Coef	<i>t</i> -Stat	dP/dX	Coef	<i>t</i> -Stat	dP/dX	
Discount (effect of interquartile change)	0.282***	15.34	3.28%	0.128***	7.99	1.51%	0.116***	6.66	1.37%	0.47%
<i>SalesRank</i>	0.065	1.42	0.76%	0.114**	2.51	1.35%	-0.065	-1.21	-0.76%	0.41%
<i>R&D</i>	-0.020***	-2.71	-0.23%	-0.019**	-2.51	-0.21%	0.145	1.15	1.71%	1.86%
<i>STD</i>	0.022**	2.44	0.27%	0.014	1.47	0.16%	-0.065	-0.62	-0.65%	0.28
<i>MarketShare</i>	-0.233***	-3.19	-2.71%	-0.273***	-3.70	-3.22%	-0.275***	-3.59	-3.24%	-3.39%
<i>Growth</i>	0.004	0.47	-0.03%	-0.007	-0.80	-0.08%	-0.004	-0.29	-0.03%	-0.01
<i>ReturnAssets</i>	-0.054***	-2.78	-0.63%	-0.125***	-6.40	-1.45%	-0.112***	-4.98	-1.31%	-1.47
<i>Leverage</i>	0.030	1.23	0.35%	0.012	0.51	0.14%	0.112***	3.65	1.32%	1.44
<i>Payout</i>	-0.000	-0.01	-0.00%	0.004	0.47	0.05%	0.006	0.67	0.07%	0.05
<i>HHFFirm</i>	0.230***	7.19	2.67%	0.233***	7.28	2.75%	0.175***	5.13	2.06%	2.36
<i>Inst</i>	0.100***	2.53	1.16%	0.080**	2.28	1.06%	0.069	1.61	0.82%	0.77
<i>HHISIC3</i>	-0.062	-1.03	-0.66%	-0.091*	-1.71	-1.07%	-0.063	-1.03	-0.74%	-0.72
<i>Amihud</i>	-0.034***	-4.05	-0.39%	-0.023***	-2.84	-0.27%	-0.027***	-2.62	-0.32%	-0.34
# obs, pseudo R^2 and all-sample frequency	100,160	0.019	6.18%	100,166	0.015	6.18%	79,100	0.018	6.24%	0.17

(continued)

Table II—Continued

Panel B: Determinants of Discount

Discount	Discount(Q)		Discount(EV/Ebids)	
	Industry	Firm	Industry	Firm
<i>Frontier</i>				
<i>SalesRank</i>	0.2818*** [13.84]	0.1185*** [5.27]	0.3419*** [18.48]	-0.0840*** [-3.77]
<i>R&D</i>	-0.0069*** [-2.81]	-0.0348*** [-12.73]	-0.7116*** [-11.82]	-1.3508*** [-19.57]
<i>ATO</i>	-0.0662*** [-15.94]	-0.0627*** [-14.62]	-0.0441*** [-9.96]	-0.0337*** [-7.64]
<i>MarketShare</i>	-0.2317*** [-7.42]	-0.1519*** [-4.64]	-0.0494* [-1.82]	-0.0088 [-0.31]
<i>Growth</i>	-0.0031*** [-17.48]	-0.0258*** [-6.67]	-0.0096*** [-14.02]	-0.0077*** [-19.29]
<i>ReturnAssets</i>	-0.2253*** [-38.61]	0.0680*** [11.24]	-0.0577*** [-3.26]	0.0364*** [5.29]
<i>Leverage</i>	-0.0063*** [-5.63]	0.0141 [1.22]	-0.0689*** [-5.39]	-0.0220** [-1.69]
<i>Payout</i>	-0.0260*** [9.71]	0.0107*** [4.51]	-0.0236*** [-4.48]	-0.0126*** [-3.68]
<i>HHFFirm</i>	-0.0291*** [-7.54]	-0.1401*** [-11.64]	-0.0093 [-0.74]	-0.0764*** [-6.32]
<i>Inst</i>	-0.1028*** [-4.41]	-0.1279*** [-5.07]	0.0235 [1.47]	-0.0533** [-3.36]
<i>HHISIC3</i>	-0.0759*** [-3.40]	-0.0372* [-1.74]	-0.1105*** [-4.84]	-0.0579** [-2.72]
<i>Amihud</i>	0.0803*** [32.84]	0.0809*** [32.27]	0.0233*** [6.88]	0.0201*** [7.40]
<i>MFFlow</i>	-0.0139*** [-8.43]	-0.0103*** [-10.13]	-0.0103*** [-4.63]	-0.0101*** [-7.13]
<i>Constant</i>	0.4649*** [28.72]	0.3268*** [18.82]	0.1037*** [9.96]	0.44618*** [28.44]
R^2	0.126	0.098	0.064	0.081
# obs	100,160	100,166	79,100	79,103

Table II, Panel A: takeover equation without feedback.

Table II, Panel B: discount equation without feedback.

Read:

- Panel A estimates takeover probability as a function of the discount without correcting for feedback.
- Discount is positive and statistically significant across all four measures.
- The economic magnitude is modest: the interquartile effect ranges from roughly 0.41 to 1.58 percentage points.
- Panel B shows that MFFlow is significantly related to the discount, confirming first-stage relevance.

Economic message: Without correcting for endogeneity, the paper would look like earlier literature: valuation matters, but only weakly. This table is the benchmark that shows why the paper needs the simultaneous system and instrument.

7.4 Main causal estimates: the trigger effect

Table III
Effects of Discount on Takeover with Feedback

This table reports results from estimating equation (9) in the 89–09 joint system. All variables are defined in Table 1. The dependent variable is Takeover. All standard errors are adjusted for heteroskedasticity and correlation clustered at the firm level. The column dP/dX gives the marginal effect on takeover probability of a one unit (or 100 percentage point) change in each regressor. Year fixed effects are used in all specifications but are unreported. The bottom of the table reports the number of observations, the pseudo R^2 , and the all-sample frequency of the dependent variable being one. Also reported are the test for weak instruments (which rejects the null that the instrumental variable is weak at any significance level higher than the indicated p -value) and the Wald test for the exogeneity of the system (which rejects the null that the joint system could be reduced to two independent equations at any significance level higher than the indicated p -value). *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels.

	Dependent Variable = Takeover							
	Discount = Discount(Q)				Discount = Discount(EV/Ebids)			
	Industry-Specific Frontier		Firm-Specific Frontier		Industry-Specific Frontier		Firm-Specific Frontier	
	Coef	t-Stat	dPrdX	t-Stat	Coef	t-Stat	dPrdX	t-Stat
Discount (effect of interquartile change)	1.371***	4.24	15.66%	0.889***	3.67	11.58%	1.512***	4.51
SubsBank	-0.259**	-2.31	-2.96%	0.002	0.02	0.02%	-0.336***	-4.34
R&D	-0.006	-0.73	-0.07%	0.014	1.13	0.17%	1.138***	4.28
ATO	0.094***	4.09	1.07%	0.0967***	3.51	0.78%	-0.067***	-3.72
MktShr	0.058	0.47	0.66%	-0.115	-1.23	-1.35%	-0.135	-1.52
Growth	0.064***	3.19	0.73%	0.017	1.49	0.20%	0.097***	3.00
BetaAsset	0.208**	2.47	2.37%	-0.170***	-7.65	-1.99%	0.000	0.00%
Leverage	0.084***	3.10	1.07%	0.001	0.04	0.01%	0.187***	6.16
Payout	0.023**	-2.16	-0.26%	-0.006	-0.07%	0.038***	3.58	0.40%
HHFirm	0.271***	8.53	3.10%	0.340***	8.11	3.38%	0.146***	3.94
Inst	0.190***	4.16	2.24%	0.195***	3.87	2.28%	0.012	0.27

(continued)

Table III—Continued

	Dependent Variable = Takeover											
	Discount = Discount(Q)				Discount = Discount(EV/Ebids)				Discount = Discount(EV/Ebids)			
	Industry-Specific Frontier		Firm-Specific Frontier		Industry-Specific Frontier		Firm-Specific Frontier		Industry-Specific Frontier		Firm-Specific Frontier	
	Coef	t-Stat	dPrdX	t-Stat	Coef	t-Stat	dPrdX	t-Stat	Coef	t-Stat	dPrdX	t-Stat
HHISCS	0.008	0.12	0.08%	-0.057	-1.07	-0.67%	0.105	1.41	1.11%	-0.007	-0.12	-0.09%
Amihud	-0.124***	-4.90	-1.42%	-0.097***	-4.03	-1.14%	-0.097***	-5.29	-0.60%	-0.050***	-4.38	-0.57%
# obs, pseudo R^2 , and all-sample frequency	100,160	0.020	6.18%	100,166	0.015	6.18%	79,100	0.018	6.24%	79,103	0.017	6.24%
Weak instrument tests												
F1, # obs and p-value	95.38 0.00				167.15 0.00				38.00 0.00			
Exogeneity tests												
Wald statistics and p-value	7.71 0.01				8.21 0.00				7.93 0.01			

Table III, continued: weak-instrument and exogeneity tests.

Table III, first part: effects of discount on takeover with feedback.

Read:

- Table III is the paper's main result.
- Once the discount is instrumented with MFFlow, the coefficient on Discount becomes much larger.
- A one-unit increase in Discount implies a marginal takeover probability effect of about 11.6% to 15.9%, depending on the specification.
- An interquartile increase in Discount raises takeover probability by about 5.7–7.6 percentage points.
- The unconditional takeover probability is about 6.2%, so the effect is economically large.

Economic message: The trigger effect is strong. The weak relation in ordinary takeover regressions is not evidence that market prices are unimportant. It is evidence that prices are endogenous and already reflect anticipated takeovers.

7.5 Feedback from takeover probability to discount

Table IV
The Feedback Effect from Takeover to Discount

This table reports the estimation of the system (8)–(9) through a regression of residual *Discount* from equation (11) on shocks to *Takeover* from equation (14). All standard errors are adjusted for heteroskedasticity and correlation double-clustering at the year and firm levels, as well as for the variation from the first-stage estimation. The associated *t*-statistics are reported in the brackets. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels.

Residual Discount	<i>Discount(Q)</i>		<i>Discount(EV/Ebitda)</i>	
	Industry	Firm	Industry	Firm
Frontier				
ξ (Shocks in <i>Takeover</i> *)	−0.163*** [−32.40]	−0.140*** [−18.66]	−0.266*** [−32.92]	−0.193*** [−33.96]
# obs	100,160	100,166	79,100	79,103
R^2	0.040	0.046	0.109	0.048

Table IV: feedback from takeover shocks to the valuation discount.

Table V
Financially Driven Takeovers

This table repeats the analyses in Table III (Panel A) and Table IV (Panel B) but only studies takeovers that are either leveraged buyouts and/or undertaken by financial sponsors. All takeovers are removed from the sample. Other *t*-statistics are reported in the brackets below the coefficient. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels.

Panel A: Effect of Discount on Takeover with Feedback				
Discount	<i>Discount(Q)</i>		<i>Discount(EV/Ebitda)</i>	
	Industry	Firm	Industry	Firm
Frontier				
Discount	1.481*** [3.96]	1.414*** [4.06]	1.471*** [3.45]	1.448*** [3.60]
(dPr/dX)	4.76% 2.29%	4.32% 2.11%	4.69% 2.21%	4.51% 2.23%
(effect of interquartile change)				
SalesRank	−0.376*** [−3.05]	−0.178* [−1.88]	−0.523*** [−3.21]	0.019 [0.24]
R&D	0.019* [1.81]	0.056*** [3.65]	0.100 [0.16]	1.094 [1.36]
ATO	0.146*** [6.28]	0.135*** [6.37]	−0.020 [−0.62]	0.095*** [6.26]
MktShr	0.187 [1.22]	0.098 [0.74]	−0.051 [−0.44]	−0.091 [−0.83]
Growth	0.088*** [3.72]	0.050*** [3.25]	0.071* [1.67]	0.074* [1.81]
BetaAsset	0.094 [0.79]	−0.320*** [−10.43]	−0.118 [−1.61]	−0.266*** [−6.58]
Leverage	0.141*** [3.25]	0.034 [0.82]	0.214*** [4.37]	0.151*** [3.10]
Payout	−0.009 [−0.54]	0.002 [0.12]	0.050*** [3.72]	0.044*** [3.56]
HHIFirm	0.126** [2.33]	0.258*** [3.72]	0.061 [1.24]	0.169*** [2.88]
Inst	0.469*** [8.03]	0.549*** [8.13]	0.234*** [2.61]	0.455*** [7.23]
HHISIC3	0.096 [1.13]	−0.031 [−0.42]	0.102 [1.03]	−0.041 [−0.51]
Amihud	−0.134*** [−4.08]	−0.132*** [−4.23]	−0.031* [−1.67]	−0.034* [−1.88]
Constant	−2.658*** [−33.31]	−2.382*** [−16.89]	−1.990*** [−6.75]	−2.465*** [−15.95]
Probability of takeover	1.38%	1.38%	1.45%	1.45%
# obs	94,802	94,808	74,901	74,904
Pseudo R^2	0.037	0.032	0.041	0.040
Exogeneity tests				
Wald statistic	6.14	7.95	4.56	5.36
<i>p</i> -value	0.01	0.00	0.03	0.02

Table V, Panel A: trigger effect for financially driven takeovers.

Read:

- Table IV estimates how takeover likelihood feeds back into the discount.
- The coefficients are negative in all four specifications and highly significant.
- The paper interprets this as evidence that higher takeover probability raises prices today, reducing the measured discount.
- Table V, Panel A repeats the trigger-effect analysis for financially driven takeovers such as LBOs and deals by financial sponsors.
- The trigger effect remains strong; the interquartile effect is about 2.1–2.3 percentage points relative to an unconditional financially driven takeover probability of roughly 1.4%.

Economic message: Table IV documents the anticipation channel. Table V shows that the main price-trigger mechanism is not only about strategic synergies; it also appears for financially driven takeovers where buying undervalued or underperforming firms is especially central.

Table V—Continued

Panel B: The Feedback Effect from Takeover to Discount				
Residual Discount	<i>Discount(Q)</i>		<i>Discount(EV/Ebitda)</i>	
Frontier	Industry	Firm	Industry	Firm
ξ (Shocks in Takeover*)	−0.272*** [−12.73]	−0.343*** [−15.39]	−0.375*** [−11.27]	−0.381*** [−12.24]
# obs	94,802	94,808	74,901	74,904
R^2	3.59%	5.80%	6.55%	6.98%

Table V, Panel B: feedback effect for financially driven takeovers.

Read:

- Panel B repeats the feedback analysis for financially driven takeovers.
- The feedback effect remains negative and statistically strong.
- The magnitude is even larger than in the full-sample feedback table for several specifications.

Economic message: Even in the subsample most closely related to financial discipline, expected takeovers feed back into prices. The market partially capitalizes the expected takeover premium before the bid, which can weaken the measured discount and potentially deter some bids.

8 How to interpret the results

8.1 Trigger effect

The trigger effect means that a lower price can make a firm more likely to be acquired. This is a real effect of financial markets because a change in market price changes the probability of a corporate-control event.

8.2 Anticipation effect

The anticipation effect means that investors understand the takeover possibility and price it in. If expected takeover probability rises, the stock price rises today, reducing the discount. The paper estimates that for the average firm, a one percentage point increase in takeover probability reduces the measured discount by about 1.2 percentage points. Since average takeover premiums are about 40%, about 0.4 percentage points can be attributed directly to next-year takeover anticipation; the rest may reflect future-year anticipation or omitted factors.

8.3 Why earlier studies found weak results

Earlier work often used raw valuation and did not solve the feedback problem. This paper shows that both choices attenuate the estimated relation:

- raw valuation is a noisy proxy for undervaluation;
- anticipated takeovers raise the current price;

- omitted synergy or technology shocks can move takeover probability and price in the same direction.

Interpretation: A weak reduced-form relation between raw valuation and takeover likelihood does not imply that prices are irrelevant. It may simply mean that the market is forward-looking.

9 Limitations and caveats

- **Instrument validity is crucial.** The exclusion restriction requires mutual fund outflows to affect takeover probability only through price pressure, not directly through target fundamentals or takeover prospects.
- **The paper estimates average effects.** The authors note that they do not have enough power to estimate how the trigger and anticipation effects vary across governance characteristics.
- **The precise mechanism is not directly observed.** Prices may matter because target shareholders use them as reference points, because acquirers and shareholders learn from them, or because takeover bargaining uses market prices as an anchor.
- **The discount measure is estimated.** Frontier valuation depends on the choice of peers, valuation metric, and quantile. The paper performs robustness checks across these choices, but the measure is still constructed.

10 Short summary

Edmans, Goldstein, and Jiang study whether stock prices affect takeover activity. They argue that the simple valuation-takeover relation is biased because prices both trigger and anticipate takeovers. To isolate exogenous price variation, they use mutual fund outflow-induced price pressure as an instrument. Their central finding is that a nonfundamental decline in valuation sharply raises takeover probability. The effect is much larger after accounting for feedback than in standard non-instrumented regressions. The paper therefore provides direct evidence that financial market prices can have real effects on corporate control.

11 Conclusion

The paper's core message is that financial markets are not merely passive predictors of real events. They can help cause real events. In the takeover market, low prices discipline managers by making firms vulnerable to acquisition. But forward-looking prices also incorporate the likelihood of future takeovers, which makes the observed discount smaller and can weaken the takeover threat. This feedback loop is the paper's key insight: prices both affect and reflect takeovers.